D03 · Shop Floor Execution & MES

03 · Real-time OEE measurement & visualization

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Real-time OEE measurement is the discipline of capturing availability, performance, and quality losses at the cell, line, and plant level — and visualizing them on shop floor dashboards that drive action within the shift, not in the monthly review. The number itself is well-known; the timing of its visibility is what separates plants that improve from plants that report.

2 WHY IT MATTERS

OEE is the most universally tracked manufacturing KPI and the most universally misunderstood. **Lagging OEE (calculated end-of-shift, reviewed end-of-week) is a measurement of what already happened**; real-time OEE is a signal for what can still be changed. The standard reference work documents the asymmetry clearly **(Nakajima, 1988 — foundational text on TPM)**: an availability loss caught within 5 minutes is a 5-minute loss; the same loss caught in the morning meeting tomorrow is an 8-hour loss. The economic argument is simple — **the cost of an OEE excursion grows linearly with detection time**, and most plants are operating with detection times measured in hours when they could be in seconds.

3 HOW IT IS DONE

1. **Define the OEE calculation at three levels — cell, line, plant — and align them.** Cell OEE is the diagnostic; line OEE is the operational; plant OEE is the leadership number. They must reconcile, but each tells a different story. Without this layered definition, the OEE conversation collapses into arguments about which number is right.
2. **Instrument the loss reasons, not just the number.** A single OEE number tells leadership the score; the 16 loss reasons tell operations what to fix. Capture availability losses by reason code (planned downtime, breakdown, setup, minor stops, etc.), performance losses by cause, and quality losses by defect category — all within the same MES capture.
3. **Visualize at the cell with a 5-second glance threshold.** The shop floor dashboard must communicate the current OEE state in a glance — green/yellow/red against shift target, current vs. previous hour, current vs. shift baseline. If the operator has to interpret a chart, the dashboard has failed.
4. **Trigger escalation automatically when OEE drops by threshold within a window.** A 12-point OEE drop within 30 minutes should fire an andon to the shift supervisor automatically. The discipline of automatic escalation prevents the silent OEE degradation that compounds across a shift.
5. **Reconcile OEE measurement weekly with engineering reality.** The OEE number drifts as equipment ages, products change, and the standard cycle time embedded in the calculation becomes wrong. Quarterly recalibration with industrial engineering keeps the number honest; without it, OEE becomes a comforting fiction.
6. **Use OEE for diagnosis, not as a KPI to incentivize.** When OEE becomes a compensation metric, the loss reasons get gamed (minor stops mysteriously disappear; reasons get reclassified as "planned"). OEE is most useful when it tells the truth — and the truth requires that nobody be punished for it.

> **WORKED EXAMPLE — ILLUSTRATIVE**
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> A food processing plant operates 7 lines and tracks OEE in monthly Excel reports. Baseline OEE = 62%, but the monthly number hides the variability: actual OEE ranges from 38% to 79% across shifts on the same line. Real-time OEE is implemented with cell-level dashboards, automatic escalation at 10-point drops, and reason codes captured by operators via touch-screen at the moment of the stop. Within 90 days, average OEE rises from 62% to **71.3%**, but the more important result is that the standard deviation across shifts drops from 14 points to 4 — meaning the floor is now executing more consistently. The financial impact is **~$2.4M annualized incremental throughput** with no capex.

IN PRACTICE

Separate diagnostic OEE from incentive metrics

Use OEE for problem-solving, kaizen events, and capex prioritization; use other metrics (schedule adherence, FPY) for individual or team performance. Conflate them and you get both dishonest OEE and unfair reviews.

Decompose into the 16 losses, not just the headline

OEE without loss decomposition is a vanity number **(Nakajima, 1988 — TPM foundational text)**. Every operator should be able to name the top three losses on their cell from memory — because the dashboards make it visible every shift.

Protect the credibility of operator-entered reason codes

The moment reason codes are used to assign blame, operators stop reporting honestly and supervisors stop trusting the data. The discipline is to firewall diagnostic use from performance-review use — explicitly, in writing.

Recalibrate the OEE definition quarterly with engineering

Theoretical cycle times drift as equipment wears and products reformulate. An OEE definition untouched for two years generates numbers that look authoritative but compare nothing to reality.

WHAT YOU WOULD USE

REAL-TIME OEE PLATFORMS

MES-native OEE (Siemens, Rockwell, AVEVA) — When the MES is already in place, native OEE modules are the path of least resistance — provided the configuration discipline holds **(Siemens — vendor; Rockwell — vendor)**.

Augury, MachineMetrics, Memex — Specialized OEE platforms that deploy faster than full MES, focused specifically on availability and performance loss capture; strong fit for plants that want OEE before committing to full MES **(MachineMetrics — vendor)**.

Power BI / Tableau dashboards on top of MES — The visualization layer that turns OEE data into shop floor screens; quality of the dashboards is determined by who designs them, not by the platform **(Microsoft — vendor; Salesforce/Tableau — vendor)**.

PLC-direct OEE counters — For simple operations — direct readout of run/stop status from the PLC into a local dashboard, no full MES required. A bridge solution, not a destination.

4 SO WHAT

The OEE that surfaces in the monthly report is the OEE that lost its chance to be improved. Define the calculation at three levels, capture loss reasons not just numbers, visualize with a 5-second glance threshold, escalate automatically on threshold drops, recalibrate quarterly with engineering, and protect the number from gaming by separating diagnostic use from incentive structures. OEE is a question, not an answer — and real-time visibility is what lets the question lead to action.

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